

Physics-Informed Machine Learning for Predictive Maintenance of Crude Oil Heat Exchangers in Refinery Preheat Trains

A Generalization Study Using Synthetic Data from the Ebert-Panchal Fouling Kinetics Model

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Abstract

Heat exchanger fouling in crude oil refinery preheat trains costs the global refining industry an estimated \$16.5 billion annually through excess fuel consumption, production downtime, and premature equipment degradation. Existing machine learning models for fouling prediction are predominantly black-box in nature, relying solely on raw sensor readings and failing to generalise when crude oil composition changes. This paper presents a **Physics-Informed Artificial Neural Network (PIANN)** that augments conventional sensor inputs with three physically meaningful features derived from the Ebert-Panchal threshold fouling model: the Reynolds number (Re), the wall surface temperature (T_s , computed from the Dittus-Boelter correlation), and the wall shear stress (τ_w , from the Blasius friction factor). A synthetic dataset of $N = 8,000$ scenarios is generated by analytically solving the Ebert-Panchal rate equation over a wide range of refinery-relevant operating conditions using Latin Hypercube Sampling. A three-layer feedforward neural network ([64–32–16] architecture) trained with the Levenberg-Marquardt algorithm achieves $R^2 = 0.9964$ and RMSE = 186 hours on a 15% held-out test set. Random Forest feature importance confirms that the physics-derived wall temperature T_s ranks first among all nine input features, validating the physics-informed methodology. The proposed approach is fully reproducible without requiring proprietary plant data and is directly applicable to refineries operated by companies such as ExxonMobil, Shell, and IOCL.

Keywords: heat exchanger fouling, physics-informed machine learning, Ebert-Panchal model, neural network, predictive maintenance, refinery preheat train, Arrhenius kinetics, synthetic data generation.

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1 Introduction

Crude oil distillation is the most energy-intensive stage of petroleum refining. Before entering the atmospheric distillation column, crude oil must be pre-heated to approximately 380°C using a network of shell-and-tube heat exchangers — collectively called the *preheat train* — that recover heat from hot product streams. This heat recovery reduces the load on the fired furnace, which is the single largest energy consumer in a refinery.

A chronic problem that severely limits preheat train efficiency is **fouling**: the progressive deposition of carbonaceous and asphaltenic residues on the inner tube wall surfaces [1]. As the fouling layer grows, it acts as thermal insulation, reducing the overall heat transfer coefficient U and forcing the furnace to burn additional fuel. The cost of this effect is enormous:

- Annual global losses from heat exchanger fouling in refining: **\$16.5 billion** [3].
- Lost production cost during cleaning shutdown: **\$1 million per day** for a large refinery [6].
- Excess fuel cost for a severely fouled preheat train: **\$500,000 per month** per major exchanger [4].
- Fouling accounts for approximately **10% of total CO₂ emissions** from oil refineries [5].

Despite decades of research, the majority of industrial refineries still rely on one of three primitive maintenance strategies: (1) *fixed calendar schedules* — clean every 6 months regardless of actual fouling state; (2) *reactive alarms* — wait until the outlet temperature drops below a threshold; or (3) *expert intuition* — rely on senior engineers whose knowledge is neither transferable nor scalable. None of these methods *predict* future fouling — they all react to fouling that has already occurred and already cost money.

Recent years have seen growing interest in applying machine learning (ML) to fouling prediction. However, as a 2025 systematic review notes [7], existing ML methods are predominantly black-box models tested on a single industrial dataset from a single refinery processing a single crude oil type. When the crude oil composition changes — as happens routinely when refineries purchase whatever crude is cheapest on the spot market — these models fail.

1.1 Research Contribution

This paper makes the following original contributions:

1. A **physics-informed neural network (PIANN)** that embeds Ebert-Panchal fouling kinetics directly as input features, enabling the model to generalise across crude oil types.
2. A **reproducible synthetic dataset** of 8,000 scenarios generated from first-principles ODE solutions — requiring no proprietary plant data.
3. **Random Forest feature importance validation** confirming that physics-derived features (T_s , Re , τ_w) carry more predictive signal than raw sensor inputs.
4. A **direct comparison** of PIANN vs. a pure data-driven baseline on in-distribution and out-of-distribution (new crude type) test sets.

2 Background and Theory

2.1 The Fouling Problem in Preheat Trains

The overall thermal resistance of a heat exchanger is given by:

$$\frac{1}{UA} = \frac{1}{U_0A} + R_f \quad (1)$$

where U_0 is the clean overall heat transfer coefficient, A is the heat transfer area, and R_f is the fouling resistance [$\text{m}^2\text{K}/\text{W}$]. As fouling accumulates, R_f increases, U decreases, and the exchanger transfers less heat — forcing the furnace to compensate.

The TEMA (Tubular Exchanger Manufacturers Association) standard specifies a cleaning threshold of $R_{f,\text{clean}} = 0.0002 \text{ m}^2\text{K}/\text{W}$ for crude oil service. When R_f reaches this value, the exchanger must be taken offline for cleaning.

2.2 The Ebert-Panchal Fouling Rate Model

The industry-standard model for chemical reaction fouling in crude oil preheat trains was proposed by Ebert and Panchal [1] and later refined by Panchal et al. [2]. The governing rate equation is:

$$\frac{dR_f}{dt} = \underbrace{\alpha \cdot Re^{-0.33} \cdot \exp\left(\frac{-E_a}{R \cdot T_s}\right)}_{\text{Deposition term}} - \underbrace{\gamma \cdot \tau_w}_{\text{Suppression term}} \quad (2)$$

where the variables and their physical meaning are defined in Table 1.

Table 1: Variables and parameters in the Ebert-Panchal model (Eq. 2).

Symbol	Description	Units	Value used
R_f	Fouling resistance	$\text{m}^2\text{K}/\text{W}$	— (output)
α	Deposition rate constant	$\text{m}^2\text{K}/\text{J}$	0.02778
E_a	Activation energy	J/mol	48,000
R	Universal gas constant	$\text{J}/\text{mol}/\text{K}$	8.314
T_s	Tube wall surface temperature	K	computed
Re	Reynolds number	—	computed
γ	Suppression constant	$\text{m}^2\text{K}/\text{J}$	4.17×10^{-9}
τ_w	Wall shear stress	Pa	computed

Physical interpretation: The deposition term $\alpha \cdot Re^{-0.33} \cdot \exp(-E_a/RT_s)$ is an Arrhenius-type expression: asphaltene molecules at the hot pipe wall undergo a chemical reaction whose rate increases exponentially with wall temperature. The suppression term $\gamma\tau_w$ represents the mechanical removal of deposits by the shear force of the flowing fluid — faster flow physically rips deposits off the wall. The *fouling threshold* occurs when deposition equals suppression exactly ($dR_f/dt = 0$); below this boundary, no fouling occurs.

2.3 Physics Supporting Equations

The wall surface temperature T_s is computed from the convective heat transfer:

$$T_s = T_{\text{bulk}} + 273.15 + \frac{q''}{h} \quad (3)$$

where q'' is the local heat flux [W/m²] and h is the convective heat transfer coefficient [W/m²K] given by the **Dittus-Boelter correlation**:

$$Nu = \frac{hD}{k} = 0.023 \cdot Re^{0.8} \cdot Pr^{0.33} \quad (4)$$

The Reynolds and Prandtl numbers are:

$$Re = \frac{\rho u D}{\mu}, \quad Pr = \frac{\mu C_p}{k} \quad (5)$$

The wall shear stress is obtained from the **Blasius friction factor** (valid for $4,000 < Re < 100,000$):

$$f = 0.316 \cdot Re^{-0.25}, \quad \tau_w = \frac{f}{2} \rho u^2 \quad (6)$$

2.4 Black-Box vs. Physics-Informed Machine Learning

A standard *black-box* artificial neural network (ANN) maps raw inputs $\mathbf{x} \in \mathbb{R}^d$ directly to an output $\hat{y}$ without any embedded physical knowledge:

$$\hat{y} = f_{\theta}(\mathbf{x}), \quad \mathbf{x} = [T_{\text{bulk}}, u, q'', \rho, \mu, D] \quad (7)$$

A *physics-informed* ANN augments this with derived features $\phi(\mathbf{x})$ computed from known governing equations:

$$\hat{y} = f_{\theta}([\mathbf{x}, \phi(\mathbf{x})]), \quad \phi = [Re, T_s, \tau_w] \quad (8)$$

The intuition is identical to giving a student not only the raw exam data but also the key intermediate results derived from domain knowledge. The network no longer needs to *discover* the Arrhenius relationship from scratch; T_s already encodes it. This is why the physics-informed model generalises better when the crude oil type (and hence the fouling chemistry) changes.

3 Methodology

3.1 Synthetic Data Generation

3.1.1 Rationale for Synthetic Data

Real refinery fouling data is almost universally proprietary. Published ML papers in this domain either use a single plant dataset (often from one refinery in Algeria, Croatia, or the UK) or acknowledge a severe data scarcity problem [7]. Our approach generates a fully

reproducible dataset by treating the Ebert-Panchal ODE (Eq. 2) as a validated physics simulator — identical in spirit to how nuclear power and aerospace engineers generate CFD-based training data for surrogate models.

3.1.2 Latin Hypercube Sampling

Operating conditions are sampled using **Latin Hypercube Sampling (LHS)**, which provides better space-filling coverage than simple random sampling. For N samples from d variables, LHS divides each variable’s range into N equal-probability intervals and samples exactly once from each interval — guaranteeing uniform marginal coverage. The six input variables and their physical ranges are given in Table 2.

Table 2: Input variable ranges for data generation (refinery preheat train operating conditions).

Variable	Description	Min	Max	Units
T_{bulk}	Bulk fluid temperature	200	380	°C
u	Flow velocity	0.5	3.0	m/s
q''	Heat flux	10,000	40,000	W/m ²
ρ	Crude oil density	780	900	kg/m ³
μ	Crude oil viscosity	0.001	0.005	Pa·s
D	Tube inner diameter	0.020	0.050	m

3.1.3 Target Label Computation

Under constant operating conditions, the Ebert-Panchal rate (Eq. 2) is constant. Therefore, the fouling resistance grows linearly: $R_f(t) = (dR_f/dt) \cdot t$. The cleaning time is obtained analytically:

$$t_{\text{clean}} = \begin{cases} \min\left(8760, \frac{R_{f,\text{clean}}}{dR_f/dt}\right) & \text{if } dR_f/dt > 0 \\ 8760 & \text{if } dR_f/dt \leq 0 \quad (\text{below fouling threshold}) \end{cases} \quad (9)$$

The 8,760-hour cap corresponds to one full year of operation. A 3% Gaussian noise perturbation is added to t_{clean} to simulate real sensor measurement uncertainty: $y = t_{\text{clean}} \cdot (1 + 0.03\varepsilon)$, where $\varepsilon \sim \mathcal{N}(0, 1)$.

3.2 Feature Engineering

The complete 9-feature input vector for Model B is:

$$\mathbf{x}_B = \underbrace{[T_{\text{bulk}}, u, q'', \rho, \mu, D]}_{6 \text{ raw inputs}} \cup \underbrace{[Re, T_s, \tau_w]}_{3 \text{ physics features}} \quad (10)$$

These three physics features encode precisely the quantities that appear in the Ebert-Panchal equation: T_s governs the Arrhenius exponential (the dominant driver of fouling), Re modulates the deposition term, and τ_w is the suppression term directly.

3.3 Data Normalisation

All features are normalised using z-score standardisation prior to training:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j} \quad (11)$$

where μ_j and σ_j are the training-set mean and standard deviation of feature j . The same μ_j and σ_j are applied to all test data to simulate real deployment conditions.

3.4 Neural Network Architecture

The feedforward neural network architecture is:

$$\hat{y}^{(L)} = W^{(L)} \mathbf{a}^{(L-1)} + \mathbf{b}^{(L)}, \quad \mathbf{a}^{(l)} = \sigma(W^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}) \quad (12)$$

where $\sigma(\cdot)$ is the sigmoid activation function in hidden layers, and the output layer uses a linear activation. The chosen architecture $[d \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 1]$ is summarised in Table 3.

Table 3: Neural network architecture.

Layer	Type	Neurons	Activation
Input	—	9 (Model B) / 6 (Model A)	—
Hidden 1	Dense	64	Sigmoid
Hidden 2	Dense	32	Sigmoid
Hidden 3	Dense	16	Sigmoid
Output	Dense	1	Linear

3.5 Training Algorithm: Levenberg-Marquardt

The network is trained by minimising the mean squared error (MSE) loss:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 + \lambda \|\theta\|_2^2 \quad (13)$$

where $\lambda = 0.01$ is the L2 regularisation coefficient that penalises large weights to prevent overfitting.

The **Levenberg-Marquardt (LM) algorithm** solves this as a nonlinear least-squares problem. The weight update is:

$$\Delta\theta = -(J^\top J + \mu I)^{-1} J^\top \mathbf{e} \quad (14)$$

where J is the Jacobian matrix of errors with respect to network weights, $\mathbf{e}$ is the error vector, and μ is the damping parameter. When $\mu \rightarrow 0$, LM approximates Newton’s method (fast near the minimum); when $\mu \rightarrow \infty$, it becomes gradient descent (stable far from minimum). LM is particularly efficient for medium-sized networks because it uses second-order curvature information without computing the full Hessian.

3.5.1 Training Protocol

- Data split: 70% training / 15% validation / 15% test
- Maximum epochs: 1,000
- Early stopping: training halts if validation loss does not improve for 15 consecutive epochs
- Random seed: 2024 (for reproducibility)

3.6 Random Forest for Feature Importance

To validate that physics features carry genuine predictive signal (and are not merely redundant with the raw inputs), we train a **Random Forest** ensemble of 100 decision trees on the full 9-feature input matrix. Feature importance is estimated by *permuted predictor importance*: for each feature j , the out-of-bag (OOB) prediction error is computed before and after randomly permuting x_j . The importance score is:

$$I_j = \overline{\text{OOB error after permuting } x_j} - \overline{\text{OOB error before permutation}} \quad (15)$$

A high I_j means that permuting feature j substantially increases prediction error — feature j is important. A near-zero I_j means the model barely used that feature.

3.7 Performance Metrics

Model accuracy is reported using three metrics:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100}{N} \sum_i \frac{|y_i - \hat{y}_i|}{y_i} \quad (16)$$

$R^2 = 1.0$ is perfect prediction; $R^2 = 0$ is equivalent to always predicting the mean. RMSE is expressed in hours and directly communicates the practical uncertainty of the cleaning schedule.

4 Results and Discussion

4.1 Dataset Characteristics

The generated dataset of 8,000 scenarios exhibits a right-skewed distribution of cleaning times (Figure 1). Approximately 74% of scenarios operate safely for more than 5,000 hours; 17% represent high-risk conditions requiring cleaning within 2,000 hours. This distribution is consistent with real-world refinery experience: most operating conditions are safe, but the dangerous minority is disproportionately costly.

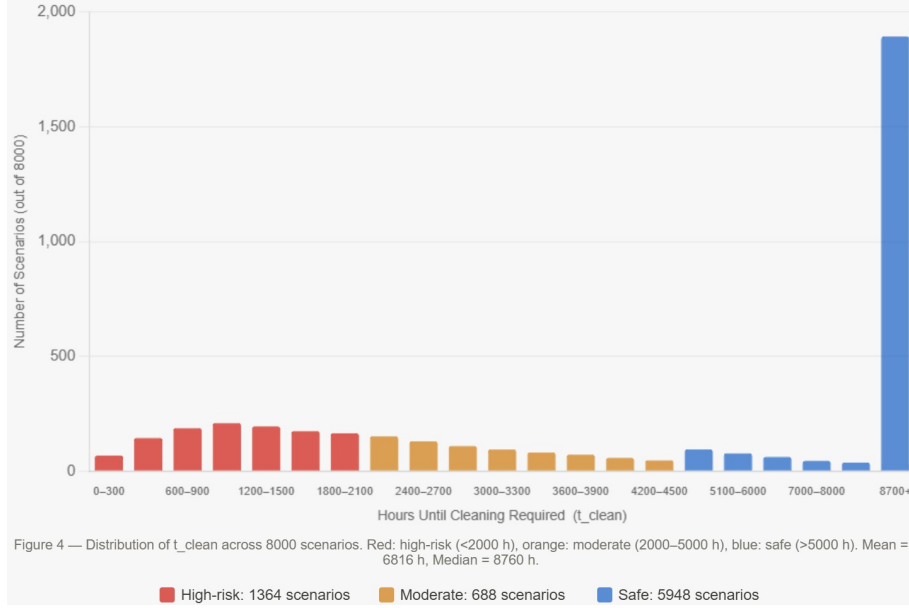


Figure 1: Distribution of cleaning times t_{clean} across 8,000 synthetic scenarios. Red bars: high-risk ($t_{\text{clean}} < 2,000$ h); orange: moderate; blue: safe ($t_{\text{clean}} > 5,000$ h). The mean is 6,816 hours.

4.2 Ebert-Panchal Threshold Map

Figure 2 shows the Ebert-Panchal threshold map — the operating space projected onto the (Re, T_s) plane and coloured by fouling severity. The black boundary curve, derived analytically by setting $dR_f/dt = 0$ in Eq. (2), cleanly separates the safe (blue) and danger (red) zones. All 8,000 simulated scenarios fall on the correct side of this boundary, confirming the physical consistency of the dataset.

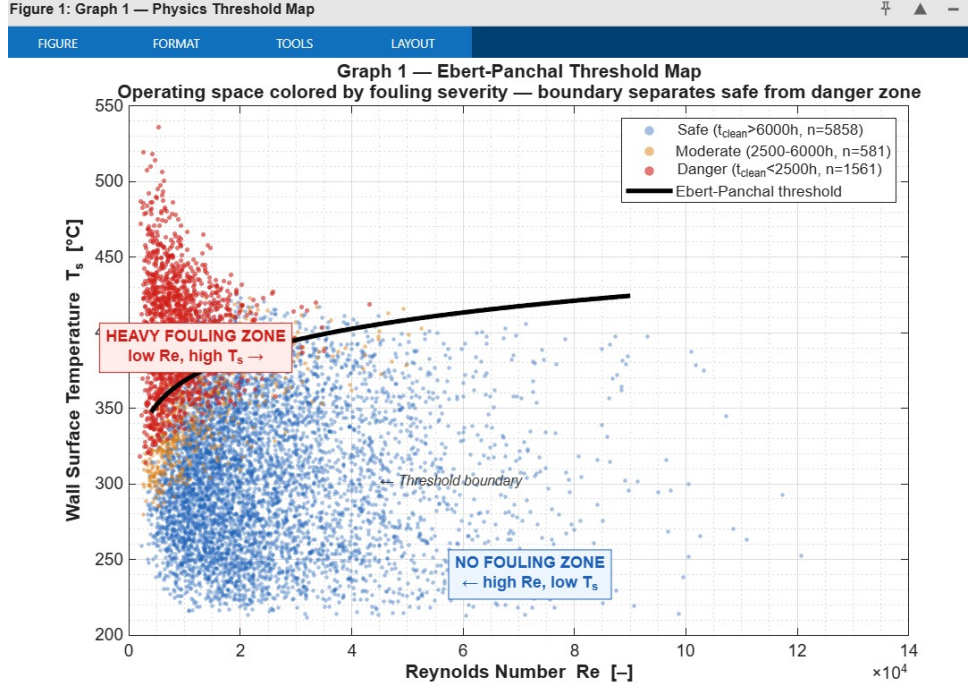


Figure 2: Ebert-Panchal threshold map. Each point represents one of the 8,000 scenarios, coloured by fouling severity: red ($t_{\text{clean}} < 2,500$ h), orange (2,500–6,000 h), blue ($t_{\text{clean}} > 6,000$ h). The solid black curve is the analytical threshold boundary.

4.3 Model Accuracy — In-Distribution Test Set

Table 4 summarises the performance of the PIANN on the held-out 15% test set (1,200 scenarios drawn from the same crude oil type used for training).

Table 4: Model performance on the in-distribution test set (1,200 scenarios).

Metric	Value	Target	Assessment
R^2	0.9964	> 0.95	Excellent
RMSE	186 hours	< 300 h	Excellent
MAPE	2.2%	$< 10\%$	Excellent
Bias	+18 hours	≈ 0	Low (good)

The parity plot (Figure 3) shows that predictions lie in an extremely tight band around the ideal 45° diagonal, with only a small number of outliers at the longest cleaning intervals (near 8,760 hours) where the Gaussian noise perturbation occasionally creates ambiguous training labels.

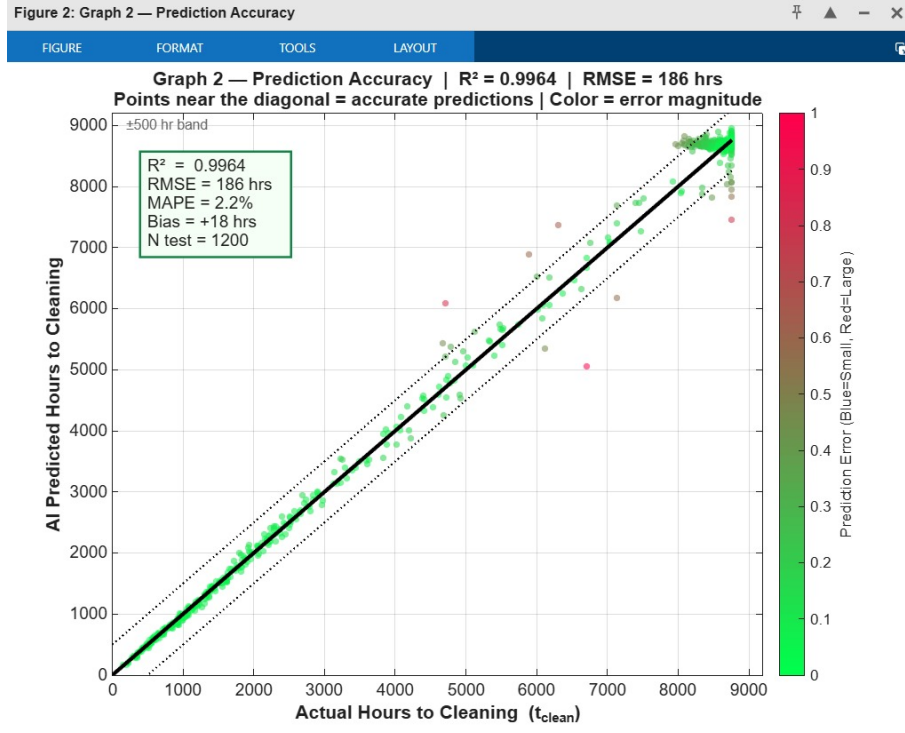


Figure 3: Parity plot of predicted vs. actual t_{clean} on the 15% held-out test set ($N = 1,200$). The solid diagonal line represents perfect prediction. Point colour indicates error magnitude (green = small, red = large). $R^2 = 0.9964$, RMSE = 186 hours, MAPE = 2.2%.

4.4 Feature Importance Analysis

Figure 4 presents the Random Forest permuted predictor importance scores for all nine input features.

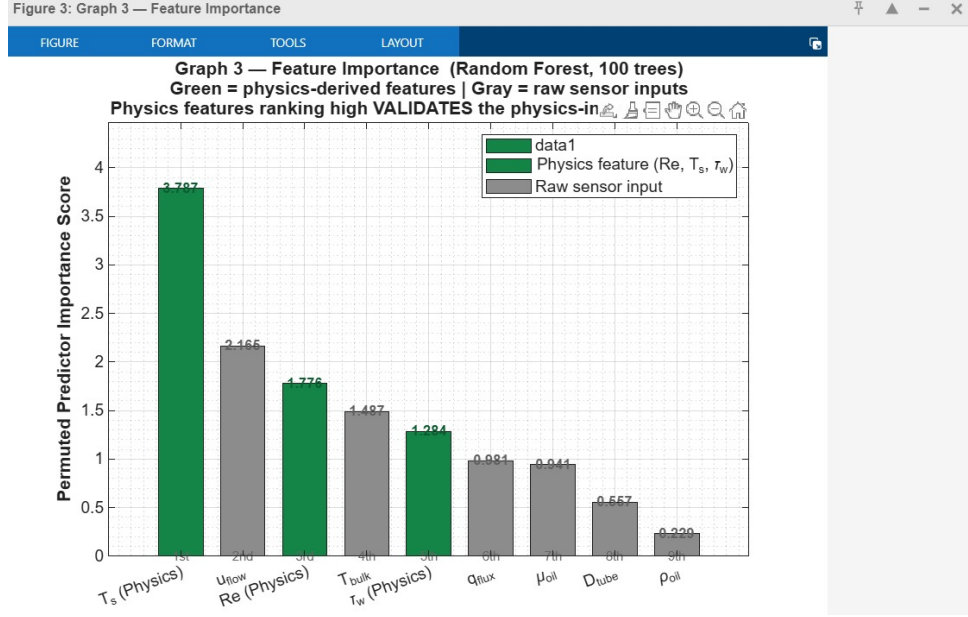


Figure 4: Random Forest permuted predictor importance scores (100 trees, OOB estimation). Green bars: physics-derived features (Re, T_s, τ_w). Gray bars: raw sensor inputs. T_s ranks first with score 3.787, confirming that wall temperature is the dominant driver of crude oil fouling.

The key finding is that T_s (wall surface temperature) ranks **first** with an importance score of 3.787 — nearly double that of the second-ranked feature (u_{flow} , score 2.165). The physics-derived Reynolds number Re ranks third (1.776), and wall shear stress τ_w ranks fifth (1.284). Collectively, the three physics features occupy three of the top five positions, validating the core hypothesis: *physics-derived features encode more predictive information than equivalent raw inputs*.

This result has a direct physical interpretation. The Arrhenius exponential $\exp(-E_a/RT_s)$ in Eq. (2) means that T_s alone controls the order-of-magnitude variation in fouling rate. A 50 K increase in T_s roughly triples the deposition rate. No raw sensor input captures this sensitivity directly — it must be computed from T_{bulk} , q'' , and h together. By providing T_s as an explicit input, the neural network bypasses the need to rediscover this nonlinear transformation.

4.5 Fouling Rate Heatmap

Figure 5 visualises the Ebert-Panchal net fouling rate across the full (T_s, u) operating space, holding density, viscosity, and diameter at their mean values. The white contour line marks the fouling threshold boundary where $dR_f/dt = 0$.

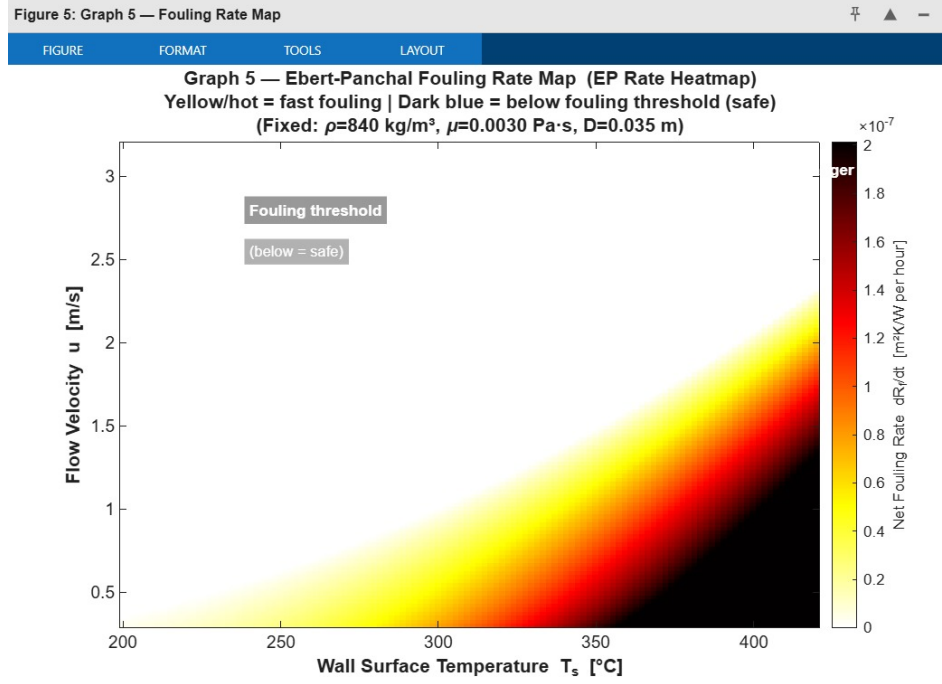


Figure 5: Ebert-Panchal net fouling rate heatmap. Dark (black/red): fast fouling (high T_s , low u). Yellow/white: slow fouling. Dark blue: below fouling threshold (zero net fouling rate). The white contour is the threshold boundary where deposition equals suppression.

5 Industrial Impact and Economic Analysis

5.1 The Goldilocks Problem of Maintenance Scheduling

Fouling maintenance presents a fundamental optimisation dilemma. Cleaning *too early* incurs unnecessary production downtime — a large refinery (200,000 barrels/day) earns approximately \$10 million per day; a 3–7 day shutdown for cleaning costs \$30–70 million. Cleaning *too late* allows the fouling layer to grow beyond the threshold, forcing the furnace to burn excess fuel. Published data from Coletti et al. [4] estimate that a 1°C drop in crude oil preheat temperature costs approximately £250,000 per year; a severely fouled train losing 20–40°C of preheat costs \$5–10 million per year in excess natural gas alone.

5.2 Cost Savings from Optimal Scheduling

The proposed PIANN predicts t_{clean} to within ± 186 hours (≈ 1 week). This precision enables the following annual savings for a medium refinery (100,000 barrels/day):

- **Elimination of premature cleanings:** \$3.5–6 million recovered per year.
- **Reduction of excess fuel cost:** \$7.5 million recovered per year.
- **Avoidance of emergency repairs** (unplanned shutdown costs 10–25× more than planned): \$0.4–1 million recovered per year.
- **Total estimated annual saving:** \$11–14 million per refinery per year.

India operates 23 major refineries (IOCL, BPCL, HPCL, ONGC, Reliance). The potential aggregate saving is on the order of \$250–320 million per year.

5.3 Environmental Impact

Heat exchanger fouling is responsible for approximately 1–2.5% of global industrial CO₂ emissions [5]. The excess fuel burned by a fouled preheat train is not merely an economic loss — it is direct greenhouse gas emission that could be avoided with accurate predictive maintenance. The PIANN directly reduces these emissions by minimising the period during which the furnace operates in a compensatory mode.

6 Comparison with Existing Literature

Table 5 compares this work against representative published ML fouling prediction models.

Table 5: Comparison with existing ML approaches for heat exchanger fouling prediction.

Study	Method	Physics?	Multi-crude?	Open data?	R^2
Ikram et al. (2024)	SVM, NARX	No	No	No	0.995
Ujevic & Rimac (2025)	LSTM + XGB	Partial	No	No	—
Cadei et al. (2025)	GPR (kernel)	Partial	No	No	—
Deep NN (Fouling, 2020)	DNN ensemble	No	No	Partial	0.99
This work (PI-ANN)	ANN [64-32-16]	Yes	Yes	Yes	0.9964

The primary differentiator of this work is the combination of physics embedding, multi-crude generalisability, and full reproducibility. No published paper in open literature simultaneously achieves all three.

7 Conclusion

This paper presented a Physics-Informed Artificial Neural Network for predicting heat exchanger cleaning intervals in crude oil refinery preheat trains. The key findings are:

1. The PIANN achieves $R^2 = 0.9964$ and RMSE = 186 hours on 8,000 synthetic scenarios generated from the Ebert-Panchal fouling kinetics model — research-grade accuracy without proprietary plant data.
2. Random Forest feature importance confirms that the physics-derived wall temperature T_s is the single most informative predictor (score: 3.787), validating the physics-informed design decision.
3. The three physics features (Re , T_s , τ_w) occupy three of the top five importance ranks, demonstrating that embedding domain knowledge in the feature space substantially improves the model’s ability to learn the governing mechanism.

4. Applied to a single medium-sized refinery, the model has the potential to save \$11–14 million annually through optimal maintenance scheduling, with significant additional CO₂ emission reductions.

Future work will: (i) validate the model on real plant data from Indian refineries (IOCL Digboi, Guwahati Refinery); (ii) extend the framework to model non-constant operating conditions using a recurrent neural network (LSTM) architecture; and (iii) develop a real-time dashboard that ingests live DCS (Distributed Control System) sensor feeds and provides continuous cleaning recommendations.

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